

Introduction To Programming

Programming Basics

- ◇ Program is a series of instructions or steps executed by the CPU.
 - ◇ go to the market.
 - ◇ then buy 4 eggs.
 - ◇ then come back to home.
- ◇ CPU don't understand our language, it just understand machine language (binary).
- ◇ Programming language => compiler => machine code.
- ◇ High level language: python, php.
- ◇ Low level language: C, assembly.

C Language

- ◇ Hello World in C:
 - ◇ `#include <stdio.h>`
 - ◇ `main() {`
 - ◇ `printf("Hello World\n");`
 - ◇ `}`
- ◇ Compile this code:
 - ◇ `gcc main.c -o myprogram`

```
student@Debian: ~  
File Edit View Search Terminal Help  
student@Debian:~$ gcc main.c -o myprogram  
main.c:2:1: warning: return type defaults to 'int' [-Wimplicit-int]  
main() {  
  ^~~~  
student@Debian:~$ ./myprogram  
Hello World  
student@Debian:~$ █
```

Data Types & Variables

◆ Data types:

- ◆ `int: x= 2; y= 5; z=1000;`
- ◆ `float: x=1.5; y= 21.6; z= 100.5;`
- ◆ `char: x="s"; y="2"; z="$";`

◆ Variables:

- ◆ `int x;`
- ◆ `float y,z;`
- ◆ `char v;`
- ◆ `x= 2; y= 15.2;`
- ◆ `z= 4.5;`
- ◆ `v= "$";`

◆ Calculations:

- ◆ `int x,y;`
- ◆ `int z;`
- ◆ `y=4;`
- ◆ `z=2;`
- ◆ `x=y+z;`
- ◆ `x=y-z;`
- ◆ `x=y*z;`
- ◆ `x=y/z;`

IF Condition

◆ If ahmed went to market then he will buy suger otherwith we will don't have any suger.

```
◆ if (ahmed == went to market)
◆ {
◆ printf("we have suger");
◆ }
◆ else
◆ {
◆ printf("we don't have suger");
◆ }
```

◆ If x value greater than 2 then i will be happy.

```
◆ int x=1;
◆ if (x > 2)
◆ {
◆ printf("I'm happy\n");
◆ }
◆ else
◆ {
◆ printf("I'm not happy\n");
◆ }
```


Loops

◇ Ahmed will say thank you five times.

◇ `printf("thank you\n");`

◇ `printf("thank you\n");`

◇ `printf("thank you\n");`

◇ `printf("thank you\n");`

◇ `printf("thank you\n");`

◇ The same code with for loop:

◇ `for (x=0; x<5; x++ )`

◇ `{`

◇ `printf("thank you\n");`

◇ `}`

Simple Program in C

- Write code to check if the student grade is greater than or equal 80% then print "you are special three times".

- `#include <stdio.h>`
- `main() {`
- `int grade=50;`
- `if (grade >= 80) {`
- `for (int x=0; x<3; x++) {`
- `printf("You are special\n");`
- `}}`
- `else {`
- `printf("You are not special\n");`
- `}}`

```
student@Debian:~$ gcc code.c -o output
code.c:2:1: warning: return type defaults to 'int' [-Wimplicit-int]
main() {
^~~~~
student@Debian:~$ ./output
You are not special
student@Debian:~$ nano code.c
student@Debian:~$ gcc code.c -o output
code.c:2:1: warning: return type defaults to 'int' [-Wimplicit-int]
main() {
^~~~~
student@Debian:~$ ./output
You are special
You are special
You are special
student@Debian:~$ █
```

Python Language

- ◇ Hello World in Python:
 - ◇ `print "Hello World"`
- ◇ Execute python code:
 - ◇ `python code.py`
- ◇ data types don't exist:
 - ◇ `x= 2`
 - ◇ `x= 9.5`
 - ◇ `x= "$"`
 - ◇ `x= "Hassan"`
- ◇ Calculations same as C language.
- ◇ If condition:
 - ◇ if x value greater than 2 then i will be happy.
 - ◇ `if x>=2:`
 - ◇ `print "happy"`
 - ◇ `else:`
 - ◇ `print "unhappy"`
- ◇ Indentation in python is very important.

Python Language

- ◊ Loops:
- ◊ Ahmed will say thank you five times.
 - ◊ `for x in range(0,3):`
 - ◊ `print "thank you"`
- ◊ Write code to check if the student grade is greater than or equal 80% then print "you are special three times".
 - ◊ `grade=50`
 - ◊ `if grade >= 80:`
 - ◊ `for counter in range(0,3):`
 - ◊ `print "You are special"`
 - ◊ `else:`
 - ◊ `print "You are not special"`
- ◊ Lists:
- ◊ `students = ["ahmed", "hassan", 12, 10.5, "$"]`
- ◊ `for student in students:`
 - ◊ `print student`
- ◊ Accessing list data:
 - ◊ `print students[0]`
 - ◊ `print students[2]`
- ◊ List functions:
 - ◊ `students.append("test")`
 - ◊ `students.remove("ahmed")`
- ◊ Take data from user:
 - ◊ `username = raw_input("enter your name: ")`

Simple Program in Python

- ◆ Write code to check if the student grade is greater than or equal 80% then print "you are special".
- ◆ `grades= [80, 50, 12, 90, 95, 100 ,0]`
- ◆ `for grade in grades:`
 - ◆ `if grade >= 80:`
 - ◆ `print "You are special your grade is:", grade`
 - ◆ `else:`
 - ◆ `print "You are not special your grade is:", grade`

```
student@Debian:~$ nano code.py
student@Debian:~$ python code.py
You are special your grade is: 80
You are not special your grade is: 50
You are not special your grade is: 12
You are special your grade is: 90
You are special your grade is: 95
You are special your grade is: 100
You are not special your grade is: 0
student@Debian:~$
```

Simple Login Program in Python

- ◇ `print "Hello friends this is my first login program in python ^_^"`

- ◇ `print "-----"`

- ◇ `credentials = ["user1:passwd1","user2:passwd2","user3:passwd3"]`

- ◇ `username = raw_input('Please enter your username: ')`

- ◇ `password = raw_input('Please enter your password: ')`

- ◇ `for credential in credentials:`

- ◇ `cred_user = credential.split(":")[0]`

- ◇ `cred_pass = credential.split(":")[1]`

- ◇ `if username == cred_user and password == cred_pass:`

- ◇ `print "You have logged in successfully ^_^"`

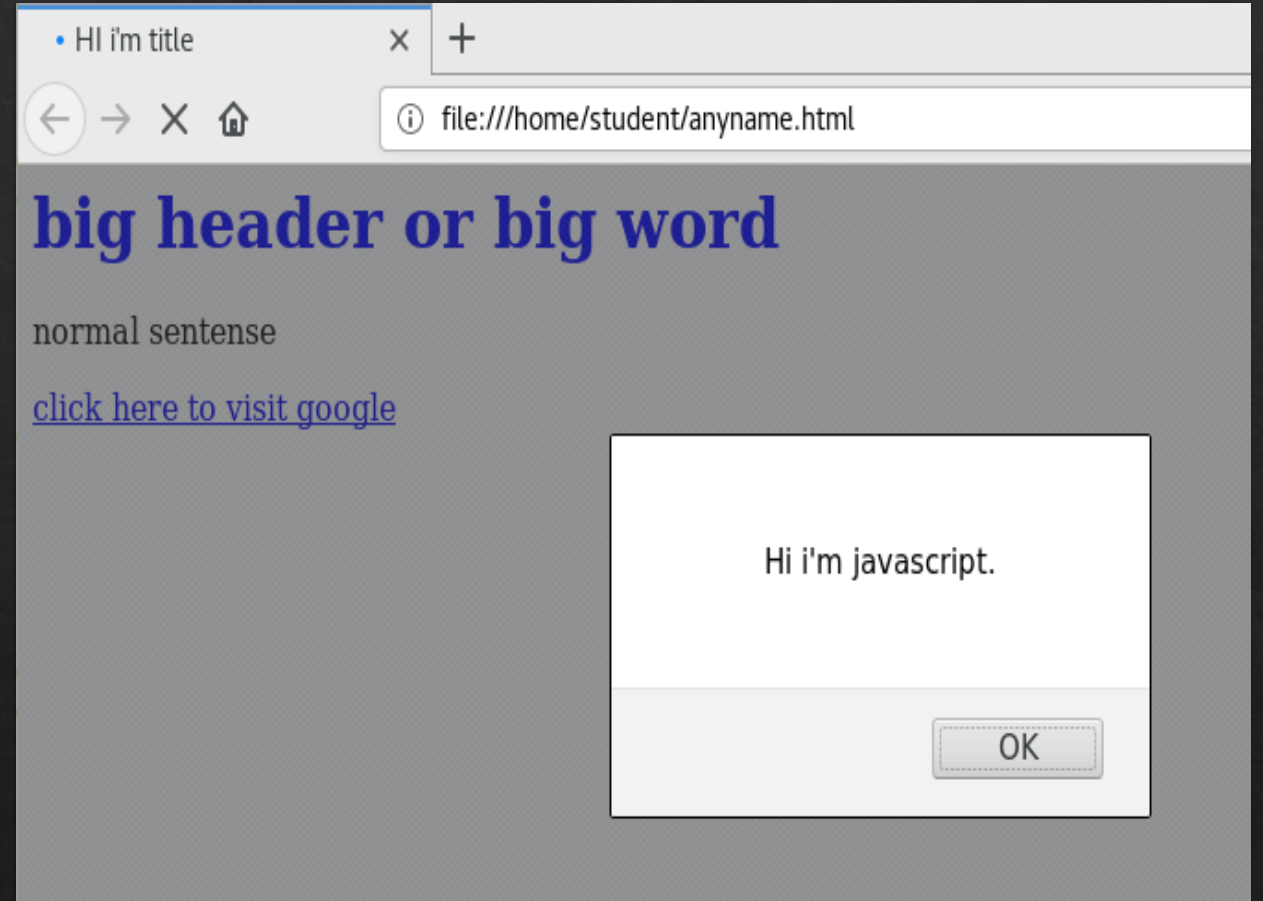
- ◇ `exit()`

- ◇ `print "Login failed, please try again."`

```
student@Debian:~$ python login.py
Hello friends this is my first login program in python ^_^
-----
Please enter your username: hassan
Please enter your password: 12345
Login failed, please try again.
student@Debian:~$ python login.py
Hello friends this is my first login program in python ^_^
-----
Please enter your username: user1
Please enter your password: passwd1
You have logged in successfully ^_^
student@Debian:~$
```

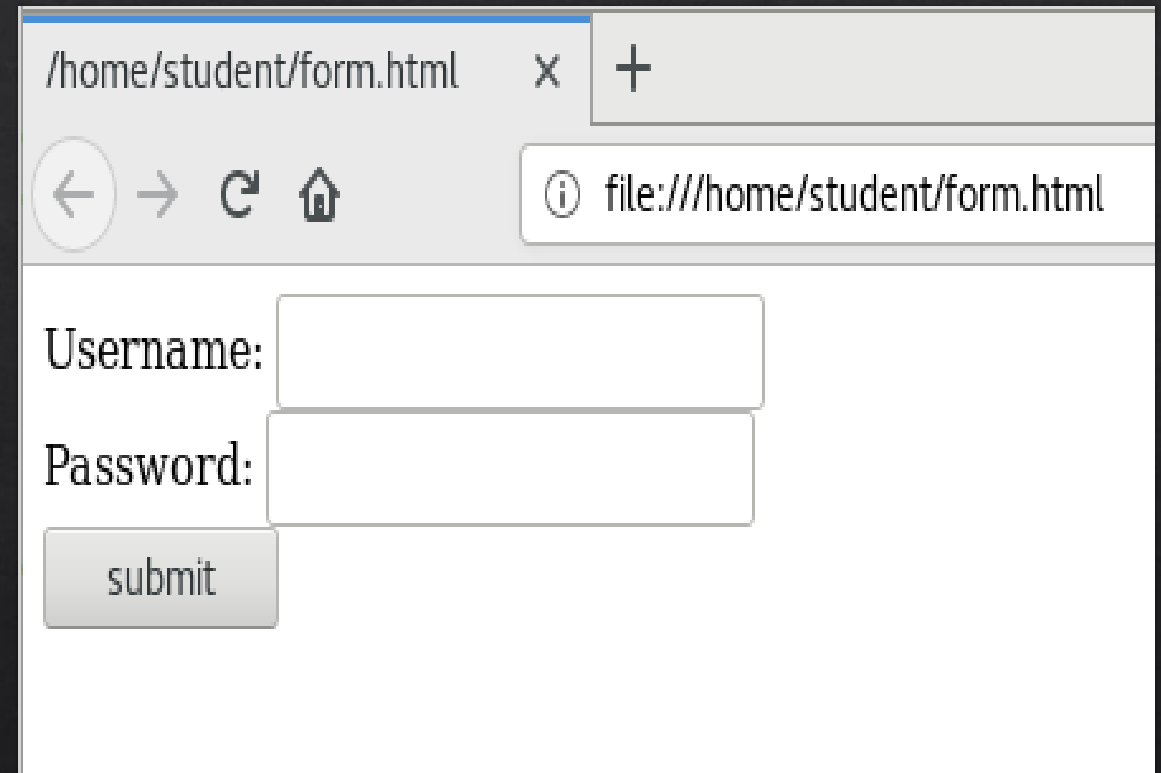
HTML

- ◇ HTML is used to define the layout and structure of the web pages.
 - ◇ `<html>`
 - ◇ `<head>`
 - ◇ `<title>HI i'm title</title>`
 - ◇ `</head>`
 - ◇ `<body>`
 - ◇ `<h1 style="color:blue">big header or big word</h1>`
 - ◇ `<p>normal sentence</p>`
 - ◇ `<a href="https://www.google.com">click here to visit google</a>`
 - ◇ `<script> alert("Hi i'm javascript.") </script>`
 - ◇ `</body>`
 - ◇ `</html>`
- ◇ save file as anyname.html



HTML Forms

- ◇ `<html>`
- ◇ `<body>`
- ◇ `<form method="post" action="code.php">`
- ◇ Username : `<input type="text" name="username"><br>`
- ◇ Password : `<input type="password" name="password"><br>`
- ◇ `<input type="submit" value="submit">`
- ◇ `</form>`
- ◇ `</body>`
- ◇ `</html>`



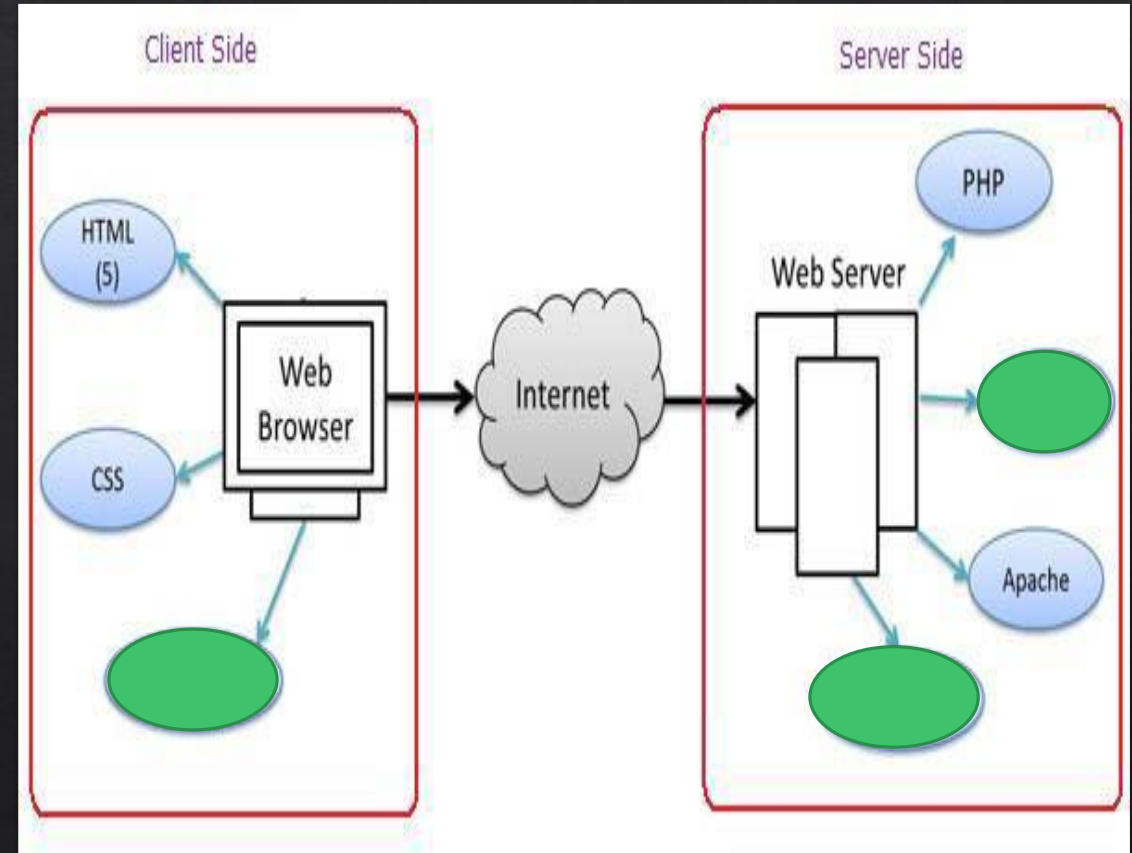
The screenshot shows a web browser window with a single tab titled `/home/student/form.html`. The address bar displays `file:///home/student/form.html`. The browser's navigation bar includes back, forward, refresh, and home icons. The rendered HTML form is displayed below the address bar. It consists of a 'Username:' label followed by a text input field, a 'Password:' label followed by a password input field, and a 'submit' button at the bottom.

Username:

Password:

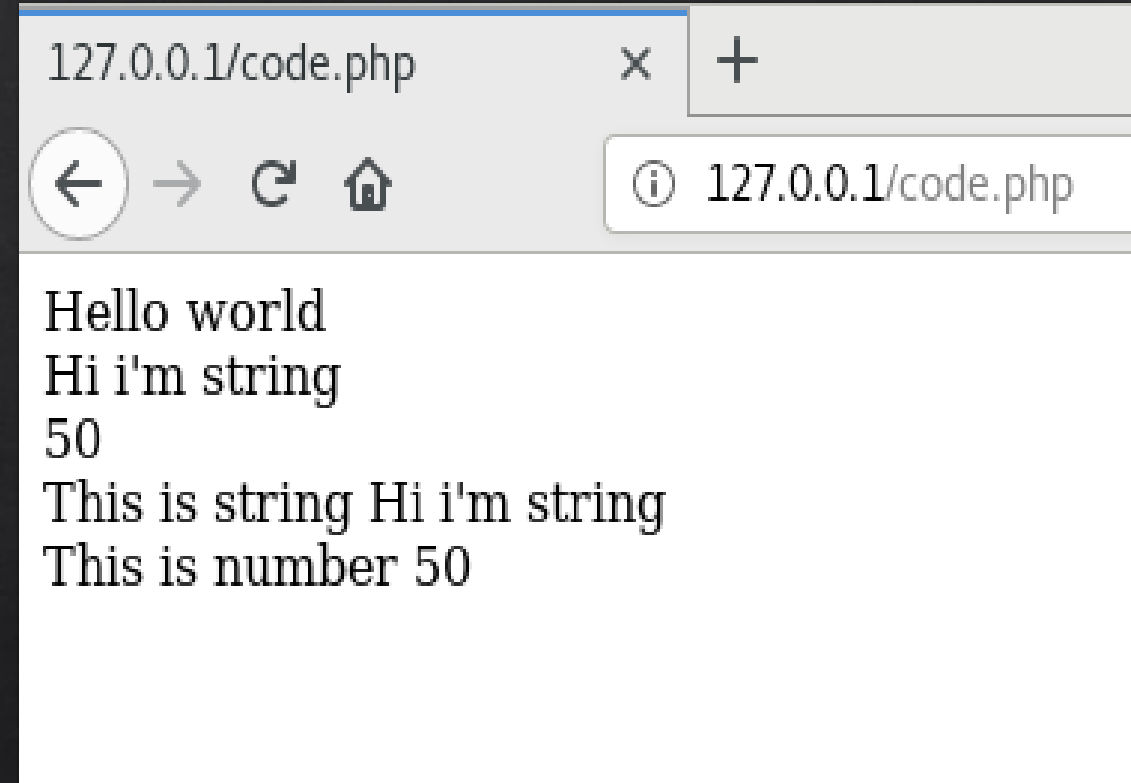
PHP Language

- ◆ To run php code you will need web server.
- ◆ Web server is a software (program) listens on TCP port 80 (HTTP), and its responsible to receive HTTP requests and send HTTP responses.
- ◆ Most common web servers apache, nginx, microsoft IIS.
- ◆ Install apache server:
 - ◆ `apt-get install apache2 php7.0`



PHP Language

- ◇ `<?php`
- ◇ `echo "Hello world <br>";`
- ◇ `$string = "Hi i'm string <br>";`
- ◇ `$number = 50;`
- ◇ `echo $string;`
- ◇ `echo $number;`
- ◇ `echo "<br> This is string $string";`
- ◇ `echo "This is number $number";`
- ◇ `?>`



IF Condition, Loops, Arrays

◆ If condition:

- ◆ `$x = 42;`
- ◆ `if ($x > 40){`
- ◆ `echo "Greater";`
- ◆ `}`
- ◆ `else {`
- ◆ `echo "Less"; }`

◆ Loops:

- ◆ `for ($i=0; $i<3; $i++){`
- ◆ `echo "Hello world"; }`

◆ Arrays == Lists in python.

- ◆ `$students = array("ahmed","hassan",12);`
- ◆ `foreach ($students as $student)`
- ◆ `{`
- ◆ `Echo $student;`
- ◆ `}`
- ◆ `echo $students[0];`
- ◆ `echo $students[1];`

Simple Login Program (HTML, PHP)

- ◇ `<html>`
- ◇ `<head>      <title>Simple Login Page</title>      </head>`
- ◇ `<body>`
- ◇ `<h1>HTML Forms</h1>`
- ◇ `<form method="post" action="code.php">`
- ◇ `Username: <input type="text" name="username"> <br>`
- ◇ `Password: <input type="password" name="password"> <br><br>`
- ◇ `<input type="submit" value="Login">`
- ◇ `</form>`
- ◇ `<p>If you click the "Login" button, the form-data will be sent to a page called "action_page.php".</p>`
- ◇ `</body>`
- ◇ `</html>`

Simple Login Program (HTML, PHP)

- ◇ <?php
- ◇ \$username = \$_POST["username"];
- ◇ \$password = \$_POST["password"];
- ◇ \$credentials = array("user1:passwd1", "user2:passwd2", "user3:passwd3");
- ◇ foreach (\$credentials as \$credential) {
- ◇ \$cred_user = explode(":", \$credential)[0];
- ◇ \$cred_pass = explode(":", \$credential)[1];
- ◇ if (\$username == \$cred_user && \$password == \$cred_pass) {
- ◇ echo "You have logged in successfully";
- ◇ exit(); } }
- ◇ echo "Login failed, please try again."; ?>